

FPGA Implementation of the Proposed DCNN Model for Detection of Tuberculosis and Pneumonia Using CXR Images

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Abstract—The automated detection of tuberculosis (TB) and pneumonia (PN) from chest X-ray (CXR) images using artificial intelligence (AI) is challenging in clinical studies for rapid diagnosis and initiation of treatment. This letter proposes the field-programmable gate array (FPGA)-based hardware implementation of a novel lightweight deep convolutional neural network (DCNN) model to detect PN and TB ailments using CXR images. Initially, the proposed DCNN (consisting of ten layers) is trained using the Google Cloud central processing unit (CPU) to obtain the model weight and bias parameters. Then, the register transfer logic (RTL) for the trained DCNN model is generated by the VIVADO high-level synthesis (HLS) framework using HLS for machine learning (HLS4ML) with fixed-point representation (8 bit for integer and 12 bit for the fractional part). The hardware implementation of the suggested DCNN model is performed using the PYNQ-Z2 FPGA framework to detect TB and PN diseases automatically. The experimental results demonstrate that the proposed DCNN model has obtained accuracy values of 96.39% and 95.63% on the Google-Cloud CPU and PYNQ-Z2 FPGA frameworks using 422 CXR images in the inference phases. The inference time of the proposed DCNN model on the PYNQ-Z2 FPGA framework is reduced by 85.19% compared to the CPU-based implementation. The suggested DCNN model has only 1831 parameters, less than the transfer learning (TFL) and existing CNN-based models to detect TB and PN using CXR images.

Index Terms—Accuracy, chest X-ray (CXR), deep convolutional neural network (DCNN), field-programmable gate array (FPGA), high-level synthesis (HLS), pneumonia (PN), PYNQ-Z2, tuberculosis (TB).

I. INTRODUCTION

THE CHEST X-ray (CXR) is used as a primary diagnostic test for the detection of different thoracic diseases due to its low cost and availability in primary healthcare centers [1]. It has been widely used to diagnose tuberculosis (TB), pneumonia (PN), and other types of pulmonary diseases [2]. The radiologists visually inspect the pathological alternations in the CXR images to detect TB and PN diseases [1]. This manual diagnosis process is cumbersome and may have false positives in diagnosing pulmonary diseases. Therefore,

computer-aided diagnostic systems based on artificial intelligence (AI) algorithms help radiologists with the automated diagnosis of TB and PN diseases. Edge computing is vital in processing medical images closer to the imaging system for faster diagnosis of different diseases by reducing latency and decreasing the need to transmit relevant patient data to centralized servers [3]. The edge devices can facilitate the deployment of privacy-preserving AI-based models without disclosing individual patient information using federated learning [4]. Hence, these devices are important in enhancing patient data privacy and security in the healthcare standard. The field-programmable gate array (FPGA) is suitable for healthcare edge computing applications due to low latency, customized hardware acceleration, parallel processing, power efficiency, and real-time processing [5]. Developing FPGA-based hardware accelerators for implementing AI algorithms is challenging for healthcare edge computing applications using CXR images to diagnose TB and PN diseases.

In recent years, various AI-based techniques have been reported in the literature to detect TB and PN diseases using CXR images [6]. Singh and Hamde [7] have computed the shape and texture features from CXR images and utilized the support vector machine (SVM)-based machine learning (ML) model for detecting TB. Similarly, the SVM classifier coupled with the shape, geometric, texture, and statistical features extracted from CXR images has been used to detect TB disease [8]. Rahman et al. [9] have compared the classification performances of nine different transfer learning (TFL)-based methods to detect TB using CXR images. The DenseNet201 model has the highest accuracy in classifying healthy and TB classes. Similarly, the TFL models, such as DenseNet-121, ResNet-18, and GoogleNet, have also been employed for PN disease detection using CXR images [10]. A CNN-based architecture with ten layers and input as CXR images has been proposed in [11] to detect PN disease. The multiscale decomposition domain deep convolutional neural network (DCNN) has been proposed by Muralidharan et al. [12] to detect PN using CXR images. Similarly, DCNN and TFL-based techniques have also been used to detect both PN and TB diseases using CXR images [13], [14].

The ML-based methods require the extraction of discriminative features, selection of relevant features, and classification algorithms to detect PN and TB diseases from CXR images [7], [8]. The deep learning methods are fully automatic

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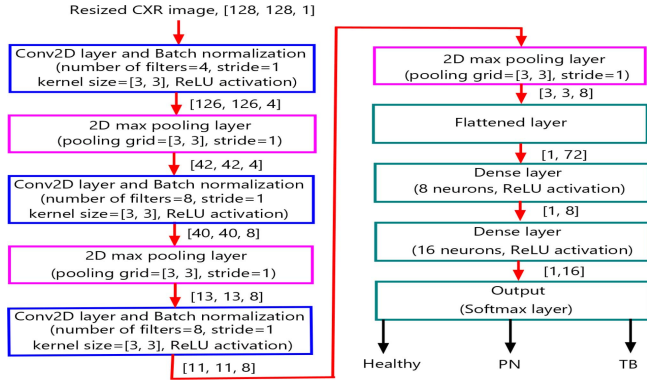


Fig. 1. Architecture of the proposed lightweight DCNN model to detect PN and TB using CXR images.

regarding the feature extraction, dimension reduction, and classification of diseases using CXR images [12]. The existing methods have not been implemented on hardware or resource-constrained edge devices to detect PN and TB diseases using CXR images. The TFL and the existing DCNN models have millions of parameters in the inference phase. Hence, these models require higher resources for hardware implementation on FPGA-based edge devices. Therefore, there is a requirement to develop a lightweight DCNN model and its hardware implementation on resource-constrained FPGA-based edge devices to detect TB and PN-based pulmonary diseases using CXR images. In this letter, we have proposed a new lightweight deep learning-based approach implemented on an FPGA device to detect PN and TB diseases using CXR images.

II. DATABASE

This letter uses two publicly available CXR image databases to assess the performance of the proposed DCNN model to detect TB and PN diseases. The first database comprises 4200 CXR images, out of which 3500 images belong to the normal class, and the remaining CXR images belong to the TB class, respectively [9]. Similarly, we have used another dataset that comprises 1525 and 4338 number of CXR images from PN and normal classes, respectively [15], [16]. From both databases, 700 CXR images from each class are used. Therefore, for normal or healthy, PN and TB classes, a total of 2100 CXR images are employed to develop and evaluate the proposed DCNN-based approach.

III. PROPOSED APPROACH

The DCNN model architecture to detect TB and PN diseases using CXR images is depicted in Fig. 1. It consists of the input layer, three 2-D convolution with batch normalization (BN) layers, three 2-D max-pooling layers, and two dense layers, followed by the output layer containing three neurons corresponding to normal (N), PN, and TB classes. The size of each CXR image in the dataset is 512×512 . In this letter, each CXR image is resized into 128×128 to implement the DCNN in both central processing unit (CPU) and PYNQ-Z2 FPGA frameworks. The resizing of the CXR image helps reduce the number of parameters and inference time of the DCNN model.

A. Proposed DCNN Model

The input to the DCNN model is the resized CXR image denoted as $\mathbf{Z} \in \mathbb{R}^{128 \times 128}$. The feature maps computed in the k th convolution layer of the proposed DCNN model are given as follows [17]:

$$\tilde{\mathbf{z}}_{q1,q2}^k = f \left[\left(\sum_{i=1}^I \sum_{j=1}^J \sum_{c=1}^C F_{i,j,c}^k \mathbf{z}_{q1+i-1,q2+j-1,c}^{k-1} \right) + \mathbf{b}^k \right] \quad (1)$$

where $F_{i,j,c}^k$ is the filter or kernel for the k th convolution layer. The $\mathbf{z}_{q1+i-1,q2+j-1,c}^{k-1}$ is the c th feature map at the $(k-1)$ th layer. The $f(\cdot)$ is the rectified linear unit (ReLU) activation function. Similarly, the feature maps of the k th pooling layer are computed as $\tilde{\mathbf{z}}_{q1,q2}^{k,c} = \text{Pooling}(\mathbf{Z}_{q1,q2}^{k-1,c})$, where $\mathbf{Z}_{q1,q2}^{k-1,c}$ is the c th feature map at the $(k-1)$ th layer. The parameters $[q1, q2]$ are the size of each feature map after the pooling layer. Moreover, the feature vector for the k th dense layer is calculated as $\tilde{\mathbf{z}}^k = f(\mathbf{W}^k \mathbf{z}^{k-1} + \mathbf{b}^k)$ [17], where $\mathbf{W}^k$, $\mathbf{b}^k$, and $\mathbf{z}^{k-1}$ are the k th dense layer weight matrix, k th dense layer bias vector, and $k-1$ th dense layer feature vector, respectively. In this letter, the fivefold cross-validation (CV) and hold-out validation schemes are employed to select training and test CXR images for the proposed DCNN model [17]. The 72%, 8%, and 20% of 2100 instances are used as training, validation, and testing in the hold-out validation. The categorical cross-entropy-based cost function and Adam optimizer are used in the training phase of the DCNN [17], [18]. Similarly, the other hyperparameters, such as the batch size of 64, the learning rate of 0.001, and the number of epochs as 20 (the early stopping with the patience of 10 epochs), are used as the training parameters. These optimal parameters are evaluated based on the grid-search technique. TFL-based models, such as ResNet50 [19], VGG19 [20], EfficientNetV2B2 [21], and MobileNet [22], are employed to detect TB and PN diseases using CXR images. The performance of these TF models is also compared with the proposed lightweight DCNN. The fine-tuning parameters in the last layer of each TF model are considered as batch size as 256, the learning rate as 0.001, the number of epochs as 2, and the Adam optimizer, respectively. The training, validation, and testing of the proposed DCNN and all TFL models are performed on the Google Cloud CPU framework.

B. FPGA Implementation of DCNN

In this letter, the quantization of the trained DCNN model is performed using HLS for ML (HLS4ML) [23] with $<20, 8>$ -based fixed-point arithmetic (8 bit for integer and $20 - 8 = 12$ bit for the fractional part) representation to detect PN and TB diseases using CXR images. Quantization helps to make the deep learning model more efficient in computation, memory usage, communication bandwidth reduction, and power consumption [23]. The fixed-point representation allows a higher range of values for the number to be represented, and it is faster and requires less memory than the floating-point representation [23]. The register transfer logic (RTL) code and post-implementation parts of the proposed DCNN model are generated using VIVADO high-level synthesis (HLS) through

TABLE I
RESULTS OF THE PROPOSED DCNN INFERENCE ON CPU AND PYNQ-Z2 FPGA PLATFORMS FOR
HOLD-OUT VALIDATION AND FIVEFOLD CV TECHNIQUES

Validation method	Implementation Platform	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Kappa
Hold-out	CPU	96.39 $\pm$ 1.73	96.46 $\pm$ 1.68	96.39 $\pm$ 1.73	96.39 $\pm$ 1.73	0.945 $\pm$ 0.026
Hold-out	PYNQ-Z2	95.63 $\pm$ 1.38	95.77 $\pm$ 1.24	95.63 $\pm$ 1.38	95.64 $\pm$ 1.37	0.934 $\pm$ 0.020
5-fold CV	CPU	95.82 $\pm$ 1.57	95.87 $\pm$ 1.54	95.82 $\pm$ 1.57	95.83 $\pm$ 1.57	0.937 $\pm$ 0.023
5-fold CV	PYNQ-Z2	95.40 $\pm$ 1.80	95.44 $\pm$ 1.78	95.40 $\pm$ 1.80	95.41 $\pm$ 1.79	0.930 $\pm$ 0.027

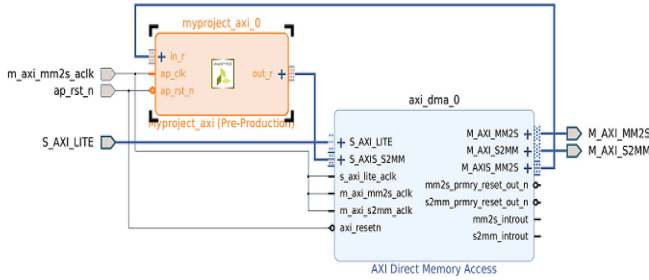


Fig. 2. PL for implementing DCNN on the PYNQ-Z2 FPGA framework.

the HLS4ML framework [23]. Then, the bit file containing the configuration and internal logic of the proposed DCNN model is obtained. Similarly, the hardware hand-off file (hwh) is generated, giving the metadata and address space of the PYNQ-Z2 FPGA device. We have used the neural network overlay in the PYNQ library to download the bit file onto the programmable logic (PL) of the FPGA framework. The RTL design of the proposed DCNN model is shown in Fig. 2. The DMA part uses an axis stream interface to transfer the CXR images in the inference phase from the programming systems (PSs) to the input part of the DCNN core. Similarly, the predicted class labels from the DCNN core are streamed to DDR memory using an axis stream interface. After obtaining the class labels, the performance measures, such as accuracy, precision, recall, F1-score, and kappa [17], are utilized to compare the software and hardware implementations of the proposed DCNN to detect TB and PN diseases using CXR images.

IV. RESULTS AND DISCUSSION

We have shown the hold-out validation and fivefold CV-based results of the proposed DCNN model to detect PN and TB diseases using CXR images in Table I. The average accuracy value is reduced from 96.39% to 95.63% when the inference is performed using the PYNQ-Z2 FPGA device to detect PM and TB diseases. Similar variations are observed in other classification measures between CPU and PYNQ-Z2 FPGA-based inferences of the proposed DCNN model to classify CXR images. The average accuracy value along all folds of the DCNN model is reduced when the inference is performed using the PYNQ-Z2 FPGA as that of the CPU framework to detect TB and PN diseases. The quantization provides the reduced precision-based representation of the weight and bias parameters of the proposed DCNN model in different layers. Due to this, there is a reduction in the accuracy value of the DCNN model inference on the PYNQ-Z2-based framework.

TABLE II
COMPARISON WITH TFR-BASED MODELS TO DETECT PN AND TB

Model	parameters	Inference time (sec)	Accuracy (%)
ResNet50 [19]	23,686,019	92.05	91.70
VGG19 [20]	20,048,963	179	95.73
EfficientNetV2B2 [21]	8,836,961	79.27	94.54
MobileNet [22]	3,278,019	28.12	87.20
Proposed DCNN	1,831	3.58	96.39

TABLE III
COMPARISON WITH EXISTING CNN METHODS TO DETECT TB AND PN

Model	12-layer CNN [13]	9-layer CNN [14]	DCNN (Proposed)
Accuracy (%)	90	92.97	96.39

The accuracy, inference time, and training parameters of the proposed DCNN model are compared with TFL methods in Table II to detect TB and PN. It is noted that the proposed DCNN has obtained an accuracy value of 96.39%, which is higher than the TFL models. The inference times for all TFR models are higher than 25 s to classify 422 CXR images on CPU. However, the inference time of the proposed DCNN to classify all 422 CXR images is only 3.58 s. The proposed DCNN model's classification accuracy value is compared with existing CNN-based techniques to detect TB and PN diseases using CXR images in Table III. The total parameters in the inference phases of both 12-layer CNN and 9-layer CNN are 26, 9, 475 and 20, 90, 6619, respectively. The accuracy values of these two models are less than the proposed DCNN to detect PN and TB ailments. The suggested DCNN model has only 1831 parameters, which is less than the TFL, 12-layer CNN, and 9-layer CNN models. Due to fewer parameters in the inference phase, the proposed DCNN model is suitable for resource-constrained edge devices for detecting TB and PN diseases using CXR images.

Moreover, we have evaluated the inference rate for processing all 422 test CXR images using the DCNN model. The inference times to process a single CXR image using the CPU-based and FPGA-based implementations of the proposed DCNN are calculated and shown in Table IV. For one CXR image, the inference time of PYNQ-Z2 FPGA implemented DCNN is 1.28 ms, much lower than the CPU-based implementation. Hence, the hardware acceleration of the proposed DCNN model occurs on the PYNQ-Z2 FPGA framework due to the faster data transfer using block random access memory (RAM). Similarly, the proposed DCNN model size has been reduced due to the conversion of the model

TABLE IV
INFERENCE TIME OF DCNN MODEL ON CPU AND FPGA

Implementation framework	Inference rate	Inference time (one image)
CPU	422 images 3.58 sec	8.50ms
PYNQ-Z2	422 images in 0.53 sec	1.28ms

TABLE V
RESOURCE UTILIZATION OF THE DCNN ON PYNQ-Z2 FPGA

Resource	Utilization	Available	Percentage of utilization
LUT	36189	53200	68.02%
LUTRAM	2084	17400	11.98%
FF	53793	106400	50.56%
BRAM	116	140	82.86%
DSP	27	220	12.27%
BUFG	1	32	3.13%

parameters from the 32-bit floating-point to fixed-point representations. The resource utilization of the proposed DCNN model implemented on the PYNQ-Z2 FPGA board is shown in Table V. It is observed that the lookup table (LUT) and block RAM utilization of the proposed DCNN model on PYNQ-Z2 FPGA are 68.02% and 82.86%, respectively. Similarly, the digital signal processing (DSP) block and flip-flop (FF) utilization on FPGA are 12.27% and 50.57%, respectively. The higher parameters-based deep learning models require more utilization of resources on FPGA. Implementing the TFL models on low-cost PYNQ-Z2 FPGA devices using HLS4ML is difficult due to limited resources (DSP, RAM, LUT, FF, and BRAM). However, the proposed DCNN is lightweight and has only 1831 total parameters. The DCNN models with $\langle 18, 8 \rangle$, $\langle 16, 8 \rangle$, $\langle 16, 6 \rangle$, and $\langle 16, 4 \rangle$ -based fixed-point representation cases require less LUT, LUTRAM, FF, BRAM, and DSP utilization as compared to $\langle 20, 8 \rangle$ case. Though the DCNN models with these low-precision-based quantization cases require lower resource utilization on PYNQ-Z2-based FPGA devices, the accuracy values of these models are less than DCNN model with $\langle 20, 8 \rangle$ -based fixed-point representation cases to detect TB and PN diseases using CXR images. The total power consumed in the PYNQ-Z2 board device to implement the proposed DCNN model is 1.929 W. Hence, the proposed DCNN model on PYNQ-Z2 FPGA can be used as an affordable and low-power resource-constrained healthcare edge computing device for real-time detection of PN and TB diseases using CXR images.

V. CONCLUSION

The FPGA implementation of the proposed lightweight DCNN model has been proposed to detect TB and PN diseases using CXR images. The trained DCNN model parameters are quantized using the 20-bit fixed-point representation. The RTL and post-synthesis implementation have been generated using VIVADO with the HLS4ML framework. The PYNQ-Z2 has been used for the FPGA-based inference of the proposed DCNN model. The DCNN model implemented on FPGA has produced an accuracy value of 95.63% in detecting PN and TB diseases. The inference time of the FPGA-implemented

DCNN is less than the CPU implementation to process CXR images for detecting TB and PN diseases.

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